# The Two Gauges of the Simplicial Frame: Zero-Sum and Standard Form, the Tropical Bridge, and the Band Structures of the Simplex Lattices

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**Abstract**

The simplicial frame describes $(N-1)$-dimensional space by $N$ coordinates modulo the gauge direction $\mathbf{1} = (1, \dots, 1)$. A *gauge* is a rule for choosing one representative per class. Papers I–IV use the **zero-sum gauge** (subtract the mean); the genesis notes use the **standard form** (subtract the minimum). We develop the second, connect the two, and test both on a family of physical lattices.

**Part I (geometry of the two gauges).** (i) *One metric.* For every representative, $\|\sum_i c_i v_i\|^2 = \frac{1}{N-1} c^\top (NI - J)c$; the genesis circle and sphere equations are its $N = 3, 4$ cases, valid in both gauges (one genesis formula for points with two zero coordinates is corrected). (ii) *A gauge is a field of probability vectors.* Every equivariant gauge has differential $I - \mathbf{1}w^\top$ with $w$ a point of the categorical simplex: the zero-sum gauge is the barycentre, the standard form the vertex field, and the temperature family is the Gibbs field. (iii) *The standard form is the tropical chart.* Its $N$ regions form the unique complete fan with the minimal number $d+1$ of cones, every two of them adjacent (the fan of $\mathbb{P}^d$, against the $2^d$ orthants of $(\mathbb{P}^1)^d$); its natural norm is the range $\max c - \min c$, whose unit ball is the Voronoi cell of the root lattice $A_{N-1}$ (hexagon, rhombic dodecahedron) with vertices exactly the $2^N - 2$ standard-form 0/1 vectors; its integer points are the weights of $SU(N)$, with dominant ones the Young diagrams and the massaction the number of removed height-$N$ columns. (iv) *The temperature bridge.* $F_\tau(c) = -\tau \log \frac{1}{N} \sum_i e^{-c_i/\tau}$ interpolates the two gauges, with $\min c \le F_\tau \le \min c + \tau \log N$, gradient the Gibbs state, and large-$\tau$ expansion $F_\tau = \bar{c} - \frac{N-1}{2N^2\tau}\|x\|^2 + \dots$ whose first correction is the metric itself.

**Part II (the simplex lattices).** The lattices whose bond star is the simplex frame — the chain ($N = 2$), honeycomb ($N = 3$), diamond ($N = 4$), and higher hyperdiamonds — are $D_N = Q \cup (Q + e_1) \subset \mathbb{Z}^N/\mathbb{Z}\mathbf{1}$, and their Brillouin zone is the torus quotient $T^N/U(1)$: the frame gauge is the Bloch sublattice phase. (v) *Nodal sets are polygon spaces.* The band touchings of $f = \sum_j t_j e^{i\theta_j}$ form the moduli space of closed planar $N$-gons with sides $t_j$, of dimension $N - 3$. For graphene the Dirac points are the two nontrivial characters of $\mathbb{Z}/3$ (the two orientations of the hopping triangle), and **the Dirac cone is the genesis circle** $l^2 + n^2 + m^2 - ln - lm - nm$. For diamond the nodal network is three circles labelled by the three opposite-edge pairs of the tetrahedron, crossing at the three $X$ points. (vi) *Grashof's rule counts nodal loops.* Under anisotropic hopping the diamond's nodal set is 0, 1, or 2 loops according to the polygon inequality and Grashof's 1883 criterion for four-bar linkages. (vii) *The tropical phase diagram.* The gapless region is the amoeba of a hyperplane and shrinks, at rate $\tau \log(N-1)$, onto the walls of the standard form; the genesis regions are the gapped phases (Kitaev's $A_x, A_y, A_z$ phases for $N = 3$). (viii) *The quantum massaction.* The Brillouin-zone average of $\log|f|$ (a Mahler measure, and for $N = 3$ the free energy of honeycomb dimers) is bounded below by the classical massaction, with equality exactly on the gapped phases; at the isotropic point it equals $\frac{3\sqrt{3}}{4\pi} L(\chi_{-3}, 2)$ for $N = 3$ and $\frac{7\zeta(3)}{2\pi^2}$ for $N = 4$. (ix) *Line graphs.* Kagome and pyrochlore carry $N-2$ flat bands whose space is "zero-sum on every simplex" (the ice rule); their pinch point is the normalized square $-\widetilde{K}(\varphi)^2/\varphi^\top M\varphi$ of Paper 01's cross-product generator.

The physics is classical (Wallace, Kasteleyn, Kitaev, Mielke, Henley, Smyth) and is reproduced, not changed. The frame removes every arbitrary choice and every $\sqrt{3}$ from the derivations, and it unifies results usually derived lattice by lattice. On the cubic lattice, where Cartesian coordinates are native, it gives nothing; the paper states this control explicitly. Section 11 designs and simulates the measurement of the one falsifiable prediction, the Grashof loop count under strain. The fastest route is a four-path interferometer, which realizes the Brillouin zone $T^4/U(1)$ synthetically, with the loops read out as vortex lines of the output phase. The real-lattice route reads the count from the number of zero-mode-free pockets on the (111) surface. All displayed identities are certified (86 + 6 checks, Appendix A).

**Keywords:** simplicial coordinates, Quadray coordinates, gauge fixing, tropical geometry, Maslov dequantization, normal fan, Voronoi cell, root lattice, Young diagram, honeycomb lattice, diamond lattice, Dirac point, nodal line, polygon space, Grashof criterion, amoeba, Mahler measure, dimer model, Kitaev

honeycomb model, kagome, pyrochlore, flat band, pinch point, spin ice.

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## 1. Introduction

### 1.1 The genesis desiderata

The genesis notes asked for a coordinate system with five properties: no negative numbers; a number of regions that grows by one per dimension, with every region adjacent to every other; products of components giving area and volume; everything Cartesian coordinates can do; and computational efficiency at least as good. Papers I–IV answered the fourth and part of the fifth in the zero-sum chart: the operators of vector calculus are conjugate to the Cartesian ones (Paper 01b, Thm. 5.3), and the finite-element matrices of the integer frame are exactly rational (Paper 01d, Thm. 4.3). The first two desiderata were never addressed, because they belong to the *other* chart. Section 4 addresses them: the standard form is nonnegative by construction, and its regions are the unique minimal complete fan, pairwise adjacent (Theorem 4.1).

### 1.2 Two gauges and a bridge

A point of the frame is a class $[c] \in \mathbb{R}^N/\mathbb{R}\mathbf{1}$. The zero-sum gauge picks $c - \bar{c}\mathbf{1}$; the standard form picks $c - (\min_i c_i)\mathbf{1}$. These two choices are the endpoints of a one-parameter family indexed by a temperature, and the family is already present in the program as the power-mean corridor of ϟ0 (its Gauge Selection

Principle), whose min-plus end the boundary paper [פ2] identified with the genesis massaction. Part I reads that corridor *geometrically on the frame*: the metric it preserves, the simplex of gauges it moves through, and the tropical geometry at its end.

### 1.3 The physics test

The assessment that preceded this paper predicted that a change of chart cannot produce new physics (Paper 01b, Thm. 5.3) and can only help where a problem permutes $N$ equivalent directions, carries a conserved total, or forces an arbitrary choice. The cleanest such problems are the lattices whose bond star *is* the simplex frame: graphene and diamond. Part II carries them into the frame. The test is pre-registered in Section 10: what counts as simpler, and a control (the cubic lattice) on which the frame is predicted to give nothing.

### 1.4 Contributions

1. **One metric for every gauge** (Theorem 2.1) and the correction of the genesis two-zero formula (Remark 2.3).
2. **Gauges as fields of probability vectors** (Theorem 3.1, Proposition 3.2), with the reading of Paper 01c's moving cubic as a Gibbs-gauge projection (Remark 3.3).
3. **The standard form as the tropical chart** (Section 4): minimal fan (Theorem 4.1), range ball = Voronoi cell of $A_{N-1}$ with 0/1 vertices (Theorem 4.3), integer points = $SU(N)$ weights and Young diagrams (Proposition 4.5).
4. **The temperature bridge** with exact bounds and the metric as first correction (Theorem 5.1).
5. **The simplex lattices** $D_N$ with Brillouin zone $T^N/U(1)$ (Section 6) and **nodal sets as polygon spaces** (Section 7): the Dirac cone as the genesis circle (Theorem 7.3), the diamond network as the three matchings of $K_4$ (Theorem 7.4), and **the Grashof count of nodal loops** (Theorem 7.5).
6. **The tropical phase diagram** (Theorem 8.1) and **the quantum massaction** (Theorem 8.2).
7. **Line graphs, the ice rule, and the pinch point** as Paper 01's cross-product square (Theorems 9.1–9.2).
8. **A scored comparison** with Cartesian derivations and a control (Section 10).

### 1.5 What is classical and what is new

Classical and cited: the power-mean corridor (ל0) and its tropical end ([פ2]); Maslov dequantization [Litvinov]; tropical hyperplanes and the fan of $\mathbb{P}^d$ [Maclagan–Sturmfels, Develin–Sturmfels, Fulton]; Voronoi cells and holes of $A_n$ [Conway–Sloane]; Young diagrams [Fulton–Harris]; graphene's band structure [Wallace, Castro Neto et al.]; Dirac-point merging [Hasegawa et al., Montambaux et al.]; the diamond nodal network [Takahashi–Murakami]; planar polygon spaces [Kapovich–Millson]; amoebas of hyperplanes [Forsberg–Passare–Tsikh]; dimers and amoebas [Kasteleyn, Kenyon–Okounkov–Sheffield]; Kitaev's phase diagram [Kitaev]; Smyth's Mahler measures [Smyth, Boyd]; random-walk moments [Borwein–Straub–Wan–Zudilin]; line-graph flat bands [Mielke, Bergman–Wu–Balents]; Coulomb-phase pinch points [Isakov et al., Henley].

New here, weighed honestly. **(a)** The geometric statements of Part I as frame facts: the gauge-invariant metric across gauges, the simplex of gauges, the minimal-fan characterization of the genesis region count, the 0/1 description of the tropical ball. These are short; their value is that they turn the genesis desiderata into theorems. **(b)** The single-chart treatment of the whole family $D_N$, in which graphene, diamond and their line graphs are one computation and the nodal sets are polygon spaces. **(c)** Three identifications we did not find in the literature, stated as such: the diamond nodal-loop count is Grashof's criterion (Theorem 7.5); the gapless region is sandwiched between the standard-form walls and their $\tau \log(N-1)$-neighbourhood, sharply (Theorem 8.1); the Coulomb-phase pinch point is $-\widetilde{K}(\varphi)^2/\varphi^\top M\varphi$ (Theorem 9.2). **(d)** The reading of the Mahler measure as a quantum correction to the genesis massaction, exact on gapped phases (Theorem 8.2). The underlying inequality is Jensen's; the reading is the contribution.

### 1.6 Notation

$\mathbf{1} = (1, \dots, 1)$, $J = \mathbf{1}\mathbf{1}^\top$, $M_N = NI - J$ (the $K_N$ graph Laplacian, Paper 01d, Thm. 2.1), $H = \{c : \sum_i c_i = 0\}$, $d = N - 1$. The unit frame is $v_i = \sqrt{N/(N-1)}\,(e_i - \frac{1}{N}\mathbf{1}) \in H$, with $\langle v_i, v_j \rangle = -\frac{1}{N-1}$ for $i \neq j$. The **massaction** of $c$ is $\mathfrak{m}(c) = \min_i c_i$ (written ך in [Genesis]); the **standard form** is $c - \mathfrak{m}(c)\mathbf{1}$. $Q = \{c \in \mathbb{Z}^N : \sum_i c_i = 0\}$ is the root lattice $A_{N-1}$. $\Delta^{N-1}$ is the open probability simplex. $\omega = e^{2\pi i/3}$.

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## Part I — The Two Gauges

## 2. One quotient, one metric

**Theorem 2.1 (The metric of the quotient).** *For every $c \in \mathbb{R}^N$,*

$$\Big\| \sum_{i=1}^{N} c_i v_i \Big\|^2 = \frac{1}{N-1}\, c^\top M_N c = \sum_i c_i^2 - \frac{2}{N-1} \sum_{i<j} c_i c_j.$$

*The right side is invariant under $c \mapsto c + k\mathbf{1}$, so it computes the same distance in every gauge.*

*Proof.* The Gram matrix of the unit frame is $\frac{N}{N-1}(I - \frac{1}{N}J) = \frac{1}{N-1}M_N$. Gauge invariance is $M_N\mathbf{1} = 0$. The second form expands $c^\top M_N c = N\sum c_i^2 - (\sum c_i)^2$. Certified for $N = 3, \dots, 6$ (G1). □

**Corollary 2.2 (The genesis circle and sphere).** *For $N = 3$, $r^2 = l^2 + n^2 + m^2 - ln - lm - nm$; for $N = 4$, $r^2 = l^2 + n^2 + m^2 + p^2 - \frac{2}{3}(ln + lm + lp + nm + np + mp)$. Both hold for every representative, in the standard form or the zero-sum gauge alike.*

The genesis notes derived these equations region by region, under the assumption that exactly one coordinate is zero, and deferred the general case ("insert a simple proof by adding $+k$"). Theorem 2.1 is that proof: the $+k$ argument is $M_N\mathbf{1} = 0$.

**Remark 2.3 (A correction to the genesis notes).** For $N = 4$ points with two zero coordinates, the genesis notes obtained by a limiting argument the separate formula $r^2 = l^2 + \frac{4}{3}p^2 - \frac{4}{3}lp$ at $(l, 0, 0, p)$. This is incorrect: Theorem 2.1 applies to that representative directly and gives $r^2 = l^2 + p^2 - \frac{2}{3}lp$. At the vertex $(0, 0, 0, p)$ the genesis formula gives $r = 2p/\sqrt{3}$, while the true distance is $p$ (G1). No separate case is needed: the one quadratic form covers every point.

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## 3. A gauge is a field of probability vectors

**Theorem 3.1 (Linear gauges).** *The linear sections of $\mathbb{R}^N \to \mathbb{R}^N/\mathbb{R}\mathbf{1}$ are in bijection with covectors $w$ satisfying $w \cdot \mathbf{1} = 1$, via $\sigma_w(c) = c - (w \cdot c)\mathbf{1}$. The map $\pi_w = I - \mathbf{1}w^\top$ is the projector with kernel $\mathbb{R}\mathbf{1}$ and image $\ker w^\top$. The positive gauges ($w_i > 0$) form the open probability simplex.*

*Proof.* A linear section is fixed by its image, a hyperplane $W$ complementary to $\mathbf{1}$; then $W = \ker w$ for a unique $w$ with $w \cdot \mathbf{1} = 1$. Idempotence, $\pi_w\mathbf{1} = 0$ and $w^\top\pi_w = 0$ are direct (G2). □

The zero-sum gauge is $w = \frac{1}{N}\mathbf{1}$, the barycentre, and it is the only orthogonal one. The space of gauges is thus the simplex itself, and the space they act on is its tangent hyperplane $H$. The frame carries both halves of the categorical simplex.

**Proposition 3.2 (Nonlinear gauges).** *Let $F : \mathbb{R}^N \to \mathbb{R}$ be equivariant, $F(c + k\mathbf{1}) = F(c) + k$. Then $\sigma_F(c) = c - F(c)\mathbf{1}$ is a section, and at points where $F$ is differentiable, $\nabla F \cdot \mathbf{1} = 1$ and $d\sigma_F = \pi_{\nabla F}$. If $F$ is nondecreasing in each coordinate, then $\nabla F$ lies in the closed simplex.*

*Proof.* Differentiate the equivariance in $k$ at $k = 0$. Monotonicity gives $\partial_i F \geq 0$. □

Three gauges organize the paper:

| gauge | $F(c)$ | $\nabla F$ | image |
|---|---|---|---|
| zero-sum (Papers I–IV) | $\bar{c} = \frac{1}{N}\sum_i c_i$ | barycentre $\frac{1}{N}\mathbf{1}$ | $H$ |
| standard form [Genesis] | $\mathfrak{m}(c) = \min_i c_i$ | vertex $e_{\arg\min c}$ (piecewise) | $\partial\mathbb{R}^N_{\geq 0}$ |
| temperature $\tau$ (Section 5) | $F_\tau(c) = -\tau \log \frac{1}{N}\sum_i e^{-c_i/\tau}$ | Gibbs state $p_\tau(c)$ | $\{\frac{1}{N}\sum_i e^{-r_i/\tau} = 1\}$ |

The standard form is the **vertex gauge**: at each point it chooses the vertex of the simplex sitting over the smallest coordinate. Its image is the boundary of the nonnegative orthant, so it is nonnegative by construction (the first genesis desideratum), and the zero coordinate names the region (the genesis "absent coordinate").

**Remark 3.3 (Paper 01c's moving cubic is a Gibbs-gauge projection).** Paper 01c, Thm. 5.1 gives $K_p^{CH}(u)v = D_p(u \odot v) - \langle u, v\rangle_{g_p} p$. Since $\langle u, v\rangle_{g_p} = \mathbf{1}^\top D_p(u \odot v)$, this is $K_p^{CH}(u)v = \pi_p^\top D_p(u \odot v)$: the Fisher-weighted componentwise product, projected onto $H$ along the gauge $w = p$. So the moving Amari–Chentsov cubic is the frozen product of Paper 01, reweighted and then gauge-fixed by the state at which it is evaluated.

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## 4. The standard form is the tropical chart

### 4.1 Regions: the minimal fan

The standard form divides the quotient into $N$ closed regions $C_i = \{[c] : c_i = \min c\}$ (zero in the $i$-th coordinate). Their union is everything; $C_i \cap C_j$ is the wall where the minimum is attained twice. In tropical language, the walls form the tropical hyperplane of $\min(c_1, \dots, c_N)$, and the regions are its linearity domains [Maclagan–Sturmfels, Develin–Sturmfels].

**Theorem 4.1 (Minimality).** *Let $\Sigma$ be a complete simplicial fan in $\mathbb{R}^d$. Then $\Sigma$ has at least $d+1$ maximal cones. Equality holds if and only if $\Sigma$ has exactly $d+1$ rays, which after positive rescaling satisfy $\sum_i u_i = 0$, and its maximal cones are the $d+1$ cones spanned by $d$-subsets of rays — the standard-form fan, up to linear isomorphism. In that case every two maximal cones share a facet.*

*Proof.* The rays positively span $\mathbb{R}^d$, so there are $n \geq d+1$ of them. Each ray $\rho$ lies in at least $d$ maximal cones: the cones containing $\rho$ project to a complete simplicial fan in $\mathbb{R}^d/\mathbb{R}\rho \cong \mathbb{R}^{d-1}$, which by induction has at least $d$ maximal cones (base: a complete fan in $\mathbb{R}^1$ has two cones, each containing one ray). Each maximal cone has $d$ rays, so counting incidences, $d \cdot \#\{\text{cones}\} \geq nd$, hence $\#\{\text{cones}\} \geq n \geq d+1$. If equality holds then $n = d+1$. Since $d+1$ vectors that span $\mathbb{R}^d$ have a one-dimensional space of linear relations, and positive spanning forces a relation with all coefficients positive, every $d$ of them are independent; rescaling makes the relation $\sum_i u_i = 0$. A complete simplicial fan on these rays has $d+1$ maximal cones, each omitting one ray, and distinct cones omit distinct rays; so the cones are exactly the omit-one cones, and two of them share the $d-1$ rays they both contain. □

**Corollary 4.2 (Projective space against a product of lines).** *The standard-form fan is the fan of the toric variety $\mathbb{P}^d$. The Cartesian orthants form the fan of $(\mathbb{P}^1)^d$: $2^d$ cones, of which only $d\,2^{d-1}$ pairs are adjacent. The genesis requirement of "one more region per dimension, every region a neighbour of every other" is satisfied exactly by the standard form and, by Theorem 4.1, by nothing else.* (Certified adjacency counts for $d = 2, 3, 4$, G4.)

The Cartesian system is a product of $d$ copies of the $N = 2$ frame (each axis is a line with two rays). The simplicial frame is a single $d$-simplex. This $\mathbb{P}^d$ versus $(\mathbb{P}^1)^d$ dichotomy reappears in Part II as the distinction between lattices whose bond star is a simplex and those whose bond star is a cross-polytope (Section 10).

### 4.2 The tropical norm and the cube corner

The natural size in the standard form is its largest coordinate. Gauge-invariantly this is the **range** $\rho(c) = \max_i c_i - \min_i c_i$, the tropical (Hilbert projective) norm.

**Theorem 4.3 (The tropical ball).** *On $H$, with the standard dot product, the unit ball $\{x \in H : \rho(x) \le 1\}$ equals the Voronoi cell of the root lattice $A_{N-1} = Q$. In standard form it is the union of the $N$ faces $\{c_i = 0\}$ of the unit cube $[0,1]^N$ — the corner of the cube at the origin. Its vertices are the $2^N - 2$ vectors $\mathbf{1}_S$ with $\emptyset \neq S \subsetneq [N]$, which are the nonzero minuscule weights (the holes of $A_{N-1}$). For $N = 3$ it is a hexagon, and for $N = 4$ a rhombic dodecahedron (14 vertices: $4 + 4 + 6$).*

*Proof.* $\rho(x) = \max_\alpha \langle \alpha, x \rangle$ over the roots $\alpha = e_i - e_j$, so the ball is $\{\langle \alpha, x \rangle \le 1\}$. The roots are the Voronoi-relevant vectors of $A_{N-1}$ and have norm 2, so this set is the Voronoi cell [Conway–Sloane, Ch. 21]. A standard form with range at most 1 is a vector in $[0,1]^N$ with a zero coordinate. A 0/1 vector $\mathbf{1}_S$ makes the $|S|(N - |S|)$ constraints $x_i - x_j \le 1$ ($i \in S$, $j \notin S$) active, and these have rank $N - 1$, so it is a vertex. Certified for $N = 3, 4, 5$: the vertex property is checked exactly, and the Voronoi identity by brute force on 3000 random points per $N$ (G5). □

**Remark 4.4 (The genesis cover image).** The genesis notes open with a picture to be read as "not a cube, not a corner": stacked cubes drawn in red, green and blue rhombi. Theorem 4.3 says what it is. Viewed along $\mathbf{1}$, the corner of the unit cube projects onto the hexagon; its three faces $\{c_i = 0\}$ project onto the three rhombi; and each rhombus is one standard-form region at unit scale. The picture *is* the standard-form chart. A surface of stacked unit cubes viewed this way is a lozenge tiling (equivalently a plane partition). Lozenge tilings are dimer coverings of the honeycomb lattice, whose band structure is Part II's first example, and their entropy is the constant of Theorem 8.2.

### 4.3 Integer points: weights and Young diagrams

**Proposition 4.5 (The integer standard form).** *(a) $\mathbb{Z}^N/\mathbb{Z}\mathbf{1}$ is the weight lattice of $SU(N)$, and the standard form gives each class a unique representative in $\mathbb{Z}^N_{\ge 0}$ with a zero coordinate. (b) $\sum_i c_i \bmod N$ (the $N$-ality) is well defined, and the root lattice $Q$ is its kernel. (c) Sorted standard forms are the partitions with at most $N - 1$ nonzero rows, i.e. the Young diagrams of the irreducible representations of $SU(N)$; for a $GL(N)$ weight $\lambda$, the massaction $\mathfrak{m}(\lambda) = \lambda_N$ counts the height-$N$ columns removed (powers of the determinant). (d) The Dynkin labels are the consecutive differences $\lambda_i - \lambda_{i+1}$, which are gauge-invariant.* (Certified for $N = 3$: N-ality, Young diagrams, and the Weyl dimensions $1, 3, \bar{3}, 8, 10$ from standard forms, G6.)

For $N = 3$ the integer standard forms make up the genesis triangular grid, which is the weight diagram of $SU(3)$, three-coloured by triality. For $N = 4$ they make up the lattice $A_3^*$ (body-centred cubic). The genesis instruction "subtract the massaction" is, on integer points, the textbook passage from $GL(N)$ to $SU(N)$: delete full columns. Part II uses the same three-colouring: the honeycomb is the triangular lattice with one triality class removed.

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## 5. The temperature bridge

**Theorem 5.1 (From mean to minimum).** *For $\tau > 0$ let $F_\tau(c) = -\tau \log \frac{1}{N} \sum_i e^{-c_i/\tau}$. Then:*

1. *(Equivariance and gradient.) $F_\tau(c + k\mathbf{1}) = F_\tau(c) + k$, and $\nabla F_\tau = p_\tau(c)$, the Gibbs state $p_{\tau,i} = e^{-c_i/\tau} / \sum_j e^{-c_j/\tau}$. So the gauge $\sigma_\tau = c - F_\tau(c)\mathbf{1}$ has differential $\pi_{p_\tau}$ (Proposition 3.2).*

2. *(Zero temperature.)* $\mathfrak{m}(c) \le F_\tau(c) \le \mathfrak{m}(c) + \tau \log N$. *Hence* $\sigma_\tau \to$ *standard form uniformly, with error at most* $\tau \log N$.
3. *(Infinite temperature.) With* $\kappa_n$ *the cumulants of the values* $c_i$ *under the uniform measure on indices,*

$$F_\tau(c) = \sum_{n \ge 1} \frac{(-1)^{n+1}\kappa_n}{n!\,\tau^{n-1}} = \bar{c} - \frac{N-1}{2N^2\tau}\Big\|\sum_i c_i v_i\Big\|^2 + \frac{\kappa_3}{6\tau^2} - \cdots,$$

*so* $\sigma_\tau \to$ *zero-sum gauge, and the first correction is the metric of Theorem 2.1.*
4. *(Unit temperature.)* $\sigma_1(c)_i = -\log(N p_{1,i})$*: the representative is the surprisal relative to the uniform distribution. Equivalently, in the coordinate* $\theta = \log p$ *the zero-sum gauge is the centred log-ratio of compositional data [Aitchison] and the e-affine chart of Paper 01c, Prop. 3.1.*

*Proof.* (1) Direct differentiation. (2) $e^{-\mathfrak{m}/\tau} \le \sum_i e^{-c_i/\tau} \le N e^{-\mathfrak{m}/\tau}$. (3) $-\tau \log \mathbb{E}[e^{-X/\tau}]$ is the cumulant generating function at $-1/\tau$; then $\kappa_2 = \frac{1}{N}\sum (c_i - \bar{c})^2 = \frac{N-1}{N^2}\|x\|^2$ by Theorem 2.1. (4) Substitute. All certified (G3; the sandwich on 2000 random cases with $N = 2, \dots, 7$). □

**Relation to the corridor.** $F_\tau(c) = -\log M_{1/\tau}(e^{-c})$, with $M_r$ the power mean. So Theorem 5.1 is ℸ0's corridor at exponent $r = 1/\tau$, written additively on the frame. Its endpoints are the geometric-mean cell ($r \to 0$, zero-sum) and the max-plus cell ($r \to \infty$, standard form), as [ℶ2] identified. The new content is items 2–3 as frame statements: the uniform error bound, and the fact that leaving the zero-sum gauge costs, to first order, exactly the squared frame distance. The limit $\tau \to 0$ is Maslov's dequantization $a \oplus_\tau b = -\tau \log(e^{-a/\tau} + e^{-b/\tau}) \to \min(a, b)$ [Litvinov].

**Physical reading.** If $c$ is a list of energies, $F_\tau + \tau \log N$ is the Helmholtz free energy $-\tau \log Z$. The massaction is the ground-state energy, and the standard form lists the excitation energies above the ground state, which is how spectra are normally quoted. The zero-sum gauge lists energies about their mean, which for a Hamiltonian means making it traceless ($\mathfrak{u}(N) \to \mathfrak{su}(N)$). The two genesis avenues are therefore the two standard ways physicists fix the zero of energy, joined by temperature.

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## Part II — The Simplex Lattices

### 6. The family $D_N$ and its Brillouin zone

**Definition 6.1 (Simplex lattices).** In $\mathbb{Z}^N/\mathbb{Z}\mathbf{1}$ let $D_N = Q \cup (Q + e_1)$: the *A sites* are the root lattice and the *B sites* its translate. Every A site $a$ is bonded to the $N$ B sites $a + e_j$; equivalently the bond star is the frame $\{v_j\}$. Realized in $H$ by $c \mapsto \sum_i c_i v_i$:

- $N = 2$: the chain;
- $N = 3$: the **honeycomb** (A and B are two of the three triality classes of the triangular lattice; the third class consists of the hexagon centres);
- $N = 4$: the **diamond** ($Q \cong A_3 \cong$ FCC, plus one class of tetrahedral holes);
- $N \ge 5$: the hyperdiamonds.

**Proposition 6.2 (Geometry and Brillouin zone).** *(a) Bonds have unit length and the Bravais lattice is* $Q$*, with nearest Bravais distance* $\sqrt{2N/(N-1)}$ *(*$\sqrt{3}$ *for the honeycomb,* $\sqrt{8/3}$ *for diamond). (b) For* $k \in H$ *put* $\theta_j = k \cdot v_j$*. Then* $\theta = \sqrt{N/(N-1)}\,k$ *and* $\sum_j \theta_j = 0$*: on the frame, the phase vector is the wavevector, rescaled. (c)* $k$ *is a reciprocal-lattice vector if and only if* $\theta \in 2\pi\mathbb{Z}^N + \mathbb{R}\mathbf{1}$*. Hence the Brillouin zone is*

$$\mathrm{BZ}(D_N) = \mathbb{R}^N/(2\pi\mathbb{Z}^N + \mathbb{R}\mathbf{1}) = T^N/U(1)_{\mathrm{diag}}.$$

*The reciprocal lattice is the weight lattice, dual to the root lattice* $Q$*.*

*Proof.* (a) $|v_i - v_j|^2 = 2 + \frac{2}{N-1}$. (b) $k \cdot (e_j - \frac{1}{N}\mathbf{1}) = k_j$ for $k \in H$. (c) The Bravais vectors are $v_i - v_j$, and $k \cdot (v_i - v_j) = \theta_i - \theta_j$, which lies in $2\pi\mathbb{Z}$ for all $i, j$ exactly when $\theta \in 2\pi\mathbb{Z}^N + \mathbb{R}\mathbf{1}$. Certified (L1, L2). □

The Brillouin zone of a simplex lattice is the compact form of the frame's own quotient: phases modulo the gauge.

**Proposition 6.3 (Tight binding).** *With hopping $t_j > 0$ on bond direction $j$, the Bloch Hamiltonian is* $\begin{pmatrix} 0 & \bar{f} \\ f & 0 \end{pmatrix}$ *with*

$$f(\theta) = \sum_{j=1}^{N} t_j e^{i\theta_j}, \qquad E = \pm|f|, \qquad |f|^2 = \sum_j t_j^2 + 2\sum_{i<j} t_i t_j \cos(\theta_i - \theta_j).$$

*(a) $f(\theta + \alpha\mathbf{1}) = e^{i\alpha} f(\theta)$: the frame gauge is the Bloch phase convention of the B sublattice, and $|f|$ is a gauge-invariant scalar field in the sense of Paper 01b, Prop. 3.1, so the group velocity $\nabla|f|$ is automatically zero-sum. (b) $|f|^2$ depends on $\theta$ only through the coboundary $d_0\theta$ on $K_N$: it is the energy of an XY model on the complete graph, and for $t_j \equiv 1$, $|f|/N$ is the Kuramoto order parameter of $N$ phases.* (Certified, L3.)

The textbook form of graphene's band, $f = 1 + e^{ik\cdot a_1} + e^{ik\cdot a_2}$, is the gauge $\theta_3 = 0$: a coordinate set to zero, as in the standard form. The symmetric form $\sum_j e^{ik\cdot\delta_j}$ is the zero-sum gauge. Physicists already move between the two gauges; the frame makes this an explicit choice that never has to be made.

**Proposition 6.4 (Moments and density of states).** *The Haar measure on $T^N/U(1)$ is that of $N$ independent uniform phases taken modulo a common rotation. Hence the band density of states of $D_N$ (with $t_j \equiv 1$) is the density of the distance travelled by an $N$-step planar random walk with unit steps in uniformly random directions [Pearson; Borwein–Straub–Wan–Zudilin]. Its moments are*

$$\langle |f|^{2n} \rangle_{\mathrm{BZ}} = \sum_{a_1+\cdots+a_N=n} \binom{n}{a_1, \ldots, a_N}^2 = \#\{\text{closed } 2n\text{-walks on } D_N\}.$$

*For the honeycomb these are $1, 3, 15, 93, 639, \ldots$; for diamond $1, 4, 28, 256, 2716, \ldots$ (the Domb numbers).* (Certified by explicit walk enumeration on $D_N$ and by exact Brillouin-zone averages, L1.)

One formula covers every lattice in the family; the per-lattice walk counts (Domb for diamond) are its specializations.

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## 7. Nodal sets are polygon spaces

**Theorem 7.1 (Nodal sets).** *The band touching $\{f = 0\} \subset T^N/U(1)$ is the moduli space of closed planar polygons with side lengths $t_1, \ldots, t_N$ modulo rotation, $\mathcal{M}(t) = \{\theta : \sum_j t_j e^{i\theta_j} = 0\}/U(1)$ [Kapovich–Millson]. It is nonempty if and only if $t_i \le \sum_{j\neq i} t_j$ for all $i$, and for generic $t$ it is a smooth closed manifold of dimension $N-3$ (or a finite set when $N = 3$).*

*Proof.* $f = 0$ says the vectors $t_j e^{i\theta_j}$ close up. Dividing by the diagonal $U(1)$ is the quotient of Proposition 6.2. The existence criterion is the polygon inequality. For the dimension, two real equations on the $(N-1)$-torus cut out a set of dimension $N-3$. □

So the dimension of the band touching is the same for every member of the family: points (Dirac cones) for $N = 3$, lines (nodal lines) for $N = 4$, surfaces for $N = 5$. It is set by the frame size, not by any lattice-specific calculation.

### 7.1 Graphene: Dirac points are characters of $\mathbb{Z}/3$

**Theorem 7.2 (Dirac points).** *(a) For $t_j \equiv 1$, in the gauge $\theta_1 = 0$, $f = 0$ exactly at $(\theta_2, \theta_3) = (\frac{2\pi}{3}, \frac{4\pi}{3})$ and $(\frac{4\pi}{3}, \frac{2\pi}{3})$. Gauge-invariantly, $\theta_j = 2\pi a j/3$ with $a = 1, 2$: the two nontrivial characters of $\mathbb{Z}/3$, and $f = 3\delta_{a,0}$. (b) $\Gamma$, $K$, $K'$ are the trivial, positive and negative characters, i.e. the zero-, positive- and*

*negative-sequence components of three-phase engineering [Fortescue]. The valley index is the orientation of the closing triangle. (c) For general $t$, the Dirac points sit at $\cos\theta_2 = (t_3^2 - t_1^2 - t_2^2)/(2t_1t_2)$ (the genesis law of cosines) and exist if and only if the $t_j$ satisfy the strict triangle inequality. The two solutions $\pm\theta$ are $K$ and $K'$; they merge when the triangle degenerates [Hasegawa et al., Montambaux et al.]. (d) With bond length 1, $|\Gamma K| = |\theta_K|/\sqrt{3/2} = 4\pi/(3\sqrt{3})$.*

*Proof.* Three unit vectors sum to zero if and only if they form an equilateral triangle. The orientation is an invariant of the triangle up to rotation; reflection $\theta \mapsto -\theta$ is time reversal. Part (c) is the law of cosines for the triangle $(t_1, t_2, t_3)$. Certified exactly (L4, L7, with 4000 random hopping triples). □

No primitive vectors are chosen, and no $\sqrt{3}$ appears until the final conversion to physical units in (d). The valley, usually introduced as "the other corner of the hexagonal zone", is here the orientation class: the datum Paper 01 shows to be essential beyond the Gram matrix (Paper 01, Prop. 2.2).

### 7.2 The Dirac cone is the genesis circle

**Theorem 7.3 (The cone).** *Let $\theta_K = (2\pi j/3)_{j\in\mathbb{Z}/3}$ and let $\varphi \in \mathbb{R}^3$ be a phase deviation. Then*

$$f(\theta_K + \varphi) = i\sum_j \omega^j \varphi_j + O(\varphi^2), \qquad \Big|\sum_j \omega^j \varphi_j\Big|^2 = \varphi_1^2 + \varphi_2^2 + \varphi_3^2 - \varphi_1\varphi_2 - \varphi_1\varphi_3 - \varphi_2\varphi_3,$$

*for every representative $\varphi$. So the constant-energy contours near a Dirac point are the genesis circles $r^2 = l^2 + n^2 + m^2 - ln - lm - nm$, and $E = \pm t\,\|\varphi\|$ in the metric of Theorem 2.1. In physical units, with $\varphi_j = a\,q\cdot v_j$,*

$$\sum_j \omega^j (q\cdot v_j) = \tfrac{3}{2}(q_x + iq_y), \qquad E = \pm\tfrac{3}{2}\,t\,a\,|q|, \qquad v_F = \tfrac{N}{N-1}\,t\,a\Big|_{N=3} = \tfrac{3}{2}ta.$$

*Proof.* Expand $e^{i\varphi_j} = 1 + i\varphi_j + O(\varphi^2)$ and use $\sum_j \omega^j = 0$. The modulus is a quadratic form invariant under $S_3$ and under the gauge, hence proportional to $\frac{1}{2}\varphi^\top M_3 \varphi$; the constant is 1. The tight-frame identity $\sum_j (q\cdot v_j)^2 = \frac{N}{N-1}|q|^2$ gives the velocity. Certified symbolically (L5). □

Three consequences. **Isotropy is automatic**: an $S_3$-invariant quadratic form on the two-dimensional $H$ is a multiple of the metric, so the cone is round without any calculation. **The velocity is a frame constant**: $\frac{3}{2}$ is the tight-frame bound $N/d$. **The Dirac operator is the triadic Wirtinger operator**: replacing $\varphi_j \mapsto -i\partial_j$ turns the low-energy Hamiltonian into $\begin{pmatrix} 0 & \bar{D} \\ D & 0 \end{pmatrix}$ with $D = \sum_j \omega^j \partial_j$, which is, up to normalization, the Wirtinger derivative written in the equilateral frame of Paper 04. Its zero modes are the triadic-holomorphic functions of that paper.

### 7.3 Diamond: three circles from the three matchings of $K_4$

**Theorem 7.4 (The diamond nodal network).** *(a) Four unit vectors sum to zero if and only if they split into two antipodal pairs. (b) For $t_j \equiv 1$, the nodal set is therefore the union of three circles*

$$C_{12|34} = \{\theta_2 = \theta_1 + \pi,\ \theta_4 = \theta_3 + \pi\}, \quad C_{13|24}, \quad C_{14|23},$$

*one for each perfect matching of $K_4$. These are the three opposite-edge pairs of the tetrahedron, i.e. the pairs exchanged by the Hodge involution $\star_E$ of Paper 01b, Thm. 10.1. (c) The circles meet pairwise in exactly one point, at $(0,0,\pi,\pi)$, $(0,\pi,0,\pi)$ and $(0,\pi,\pi,0)$ modulo gauge. These are the three $X$ points: $|\theta_X|/\sqrt{4/3} = \pi\sqrt{3}/2 = |\Gamma X|$ for bond length 1. (d) Reduced to the first Brillouin zone, each circle has $|k|$ ranging exactly over $[|\Gamma X|, |\Gamma W|]$, with minimum at its two $X$ points, which is consistent with the known $X$–$W$ nodal lines [Takahashi–Murakami].*

*Proof.* (a) If $u_1 + u_2 = w \neq 0$, the unordered pair of unit vectors with sum $w$ is unique, so $\{u_3, u_4\} = \{-u_1, -u_2\}$. If $w = 0$, then $u_2 = -u_1$ and $u_4 = -u_3$. (b) follows, with the labelling as stated. (c) Two matchings together force a unique configuration up to gauge. (d) is certified numerically by reducing 241 points of $C_{12|34}$ to the Voronoi cell of the reciprocal lattice (L6); that the circle lies *on* the zone boundary is consistent with the literature but is not separately certified here. □

The structure of the diamond nodal network — three loops crossing pairwise at three points — is the statement that $K_4$ has three perfect matchings. The equilateral planar quadrilateral space (three circles pairwise touching [Kapovich–Millson]) is the same object. The three loops are the same three opposite-edge pairs that give $\star_E$, the three independent cycles $b_1(K_4) = 3$ (Paper 01d, Sec. 7), and the three rotation axes.

### 7.4 Anisotropy: Grashof's rule counts the nodal loops

**Theorem 7.5 (Nodal loops of the anisotropic diamond).** *Let the four hoppings, sorted, be $s \leq p \leq q \leq l$, and assume no equality in what follows. The nodal set of $D_4$ is*

- *empty (gapped) if $l > s + p + q$;*
- *one loop if $l < s + p + q$ and $s + l > p + q$ (non-Grashof);*
- *two loops if $s + l < p + q$ (Grashof).*

*On the walls $s + l = p + q$ the loops touch: these are the change-point linkages, and the isotropic diamond lies on all of them at once.*

*Proof.* Gauge-fix and pair the longest and shortest hoppings: $w(\alpha) = t_l + t_s e^{i\alpha}$, so $|w|$ sweeps $[l - s,\ l + s]$ as $\alpha$ goes around, monotonically on each half-turn. For fixed $\alpha$ the remaining two vectors close the polygon if and only if $|w| \in [|p - q|,\ p + q]$. When this holds strictly there are exactly two solutions (the two orientations of the triangle with sides $|w|, p, q$), and they merge at the endpoints. Since $s \leq p, q \leq l$, we always have $|p - q| \leq l - s$: the lower constraint never binds, so the allowed set of $\alpha$ is $\{|w(\alpha)| \leq p + q\}$. If $p + q \geq l + s$ this is the whole circle, and the two orientation branches never merge (the triangle never degenerates), giving two disjoint loops. If $l - s < p + q < l + s$ the allowed set is one arc around $\alpha = \pi$, at whose ends the branches merge, giving one loop. If $p + q < l - s$, i.e. $l > s + p + q$, the set is empty. Certified: the four-bar count agrees with an independent count (thresholding $|f|$ on a $448^3$ grid of the Brillouin zone, with periodic connected-component labelling) on 56/56 random anisotropic diamonds, and with Grashof's criterion on 56/56 (L8). □

The configuration space of a four-bar linkage in the plane, modulo rotation, is the Brillouin-zone nodal set of an anisotropic diamond, and Grashof's 1883 criterion for whether a link can rotate fully is the criterion for whether the nodal network splits into two loops or one. We did not find this identification in the literature on anisotropic diamond models [Takahashi–Murakami].

**Corollary 7.6 (Strain).** *In the nearest-neighbour model, uniaxial strain changes the hoppings as follows.*

- *Along* [001] *all four bonds stay equivalent: isotropic.*
- *Along* [110] *the bonds pair up, $t = (a, a, b, b)$: $s + l = p + q$, so the system stays on a Grashof wall and the loops keep touching.*
- *Along* [111] *one bond is singled out, $t = (t', 1, 1, 1)$: two loops for $t' < 1$, one loop for $1 < t' < 3$, gapped for $t' > 3$ (L8 examples).*

The [111] distortion is the one Fu, Kane and Mele used to make a strong topological insulator from the spin–orbit-coupled diamond model [Fu–Kane–Mele]. Corollary 7.6 describes the spin–orbit-free nodal topology under the same distortion.

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## 8. The tropical phase diagram and the quantum massaction

Write the hoppings in the standard form of Part I: $t_j = e^{-c_j/\tau}$. The dominant hopping is the minimal $c_j$, so the strongest bond is the massaction coordinate, and $\tau$ is the Maslov temperature of Section 5.

### 8.1 The gapless region shrinks to the standard-form walls

**Theorem 8.1 (Tropical phase diagram).** *Let $G_\tau = \{c : e^{-c_i/\tau} \le \sum_{j\neq i} e^{-c_j/\tau}\ \forall i\}$ be the gapless region. It is $-\tau$ times the amoeba of the hyperplane $\{\sum_j z_j = 0\}$ [Forsberg–Passare–Tsikh]. Let $W$ be the standard-form walls (minimum attained at least twice) and $\delta(c)$ the gap between the two smallest coordinates (the smallest nonzero standard-form coordinate when the minimum is unique). Then*

$$W \subseteq G_\tau \subseteq \{c : \delta(c) \le \tau \log(N-1)\},$$

*and the right inclusion is sharp: $(0, \delta, \dots, \delta)$ lies on $\partial G_\tau$ exactly when $\delta = \tau\log(N-1)$. As $\tau \to 0$ the gapless region collapses onto the tropical hyperplane, and the interiors of the $N$ standard-form regions become the $N$ gapped phases, each labelled by its dominant bond.*

*Proof.* On a wall two coordinates tie for the minimum, and every polygon inequality holds. Off the walls, if $c_i$ is the unique minimum then $\sum_{j\neq i} e^{-c_j/\tau} \le (N-1)e^{-(c_i+\delta)/\tau}$, which is less than $e^{-c_i/\tau}$ once $\delta > \tau\log(N-1)$. Certified on 20000 random cases for $N = 3, \dots, 6$, with sharpness for $N = 3, \dots, 8$ (L9). □

For $N = 3$ this is the phase diagram of Kitaev's honeycomb model, whose Majorana spectrum is $\pm|f|$ with $t_j = |J_j|$ [Kitaev]. Kitaev draws it on the simplex $\sum_j t_j = 1$ (the linear chart), where the gapless B phase is the medial triangle $\max_j t_j \le \frac{1}{2}$ (L9) and the three corners are the gapped toric-code phases $A_x, A_y, A_z$. In the logarithmic chart the B phase is the amoeba; at zero temperature it is the tropical line, and the $A$ phases are the three genesis regions. The genesis regions, labelled by the absent coordinate, are the gapped phases labelled by the dominant bond. In the dimer reading they are the frozen phases, in which every lozenge has the same orientation [Kenyon–Okounkov–Sheffield].

### 8.2 The Mahler measure is the quantum massaction

**Theorem 8.2 (Quantum massaction).** *Let $R(t) = \int_{T^N/U(1)} \log|\sum_j t_j e^{i\theta_j}|\, d\theta$, the Mahler measure $m(t_1 + t_2x_2 + \cdots + t_Nx_N)$. Then:*

1. $-\log\max_j t_j \ \ge\ -R(t)\ \ge\ -\log\sum_j t_j$; *in the variables $t_j = e^{-c_j}$, $-R \le \mathfrak{m}(c)$.*
2. *Equality $R(t) = \log\max_j t_j$ holds if and only if $t_{\max} \ge \sum_{others} t_j$: the quantum and classical massactions agree exactly on the closed gapped phases, and the excess $R(t) - \log t_{\max} > 0$ is a gapless-phase order parameter.*
3. *(Sandwich.) With $R_\tau(c) = \tau R(e^{-c/\tau})$: $-R_\tau(c) \le \mathfrak{m}(c) \le F_\tau(c)$, each within $\tau\log N$. The massaction lies between the band-averaged free energy ($-R_\tau$) and the band-top free energy ($F_\tau = -\tau\log\frac{1}{N}|f(\Gamma)|$).*
4. *(Isotropic values.) $R = 0$ for $N = 2$;*

$$R = m(1+x+y) = \frac{3\sqrt{3}}{4\pi}L(\chi_{-3}, 2) = 0.3230659472\dots \quad (N=3),$$
$$R = m(1+x+y+z) = \frac{7\zeta(3)}{2\pi^2} = 0.4262783988\dots \quad (N=4)$$

   *[Smyth, Boyd].*
5. *(Meaning.) $R$ is the Brillouin-zone average of $\log|E|$, i.e. $\frac{1}{2}\log\det$ of the hopping operator per unit cell. For $N = 3$ (planar, no Kasteleyn signs needed) it is the free energy per site of the honeycomb dimer model with weights $t$ [Kasteleyn, Kenyon–Okounkov–Sheffield], and its isotropic value is the entropy of lozenge tilings. For $N = 4$ it does not count dimers.*

*Proof.* (1)–(2): integrate first over the phase of the largest $t_i$. Jensen's formula gives $\int \log|t_i e^{i\theta_i} + C| \frac{d\theta_i}{2\pi} = \log\max(t_i, |C|) \geq \log t_i$, with equality almost everywhere if and only if $|C| \leq t_i$ for almost every remaining phase, i.e. $\sum_{j\neq i} t_j \leq t_i$. The upper bound is $|f| \leq \sum_j t_j$. (3) combines (1) with Theorem 5.1(2). (4) are Smyth's evaluations; certified to 25 and 9 digits respectively (L10), with (2) checked on gapped, boundary, gapless and isotropic hoppings and (3) on 40 random cases. □

The genesis notes coined "massaction" with the gloss "mass action time energy?". Theorem 8.2 gives it a precise physical meaning: the massaction is the zero-temperature free energy of the frame; its quantum correction vanishes exactly on the gapped phases; and at the isotropic honeycomb point that correction is an $L$-value of $\chi_{-3}$, the program's chiral character mod 3. At diamond it is $\zeta(3)$. *Fence:* these are classical evaluations by Smyth; we claim only that they appear here, as the quantum massaction of the frame, and nothing about the program's arithmetic questions follows.

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## 9. Line graphs: kagome, pyrochlore, and the ice rule

The kagome and pyrochlore lattices are the line graphs of the honeycomb and diamond lattices: their sites are the bonds of $D_N$, $N$ per cell, one per frame direction.

**Theorem 9.1 (Flat bands are the zero-sum condition).** *With unit hopping, the Bloch Hamiltonian of the line graph of $D_N$ is $H_{LG}(\theta) = B^\dagger B - 2I$, where $B = \begin{pmatrix} \mathbf{1}^\top \\ (e^{i\theta_j})_j \end{pmatrix}$. Its bands are $N-2$ flat bands at $-2$ and two dispersive bands $N - 2 \pm |f(\theta)|$, which touch the flat bands at $\Gamma$. The flat-band space at $\theta$ is $\ker B = \{v : \sum_j v_j = 0,\ \sum_j e^{i\theta_j} v_j = 0\}$: zero sum on the up-simplex and on the down-simplex. At $\Gamma$ it is the whole zero-sum hyperplane $H$.*

*Proof.* Two bonds are adjacent if they share an A site (same cell, amplitude 1) or a B site (cells differing by $e_i - e_j$, phase $e^{i(\theta_j - \theta_i)}$), which gives $\mathbf{1}\mathbf{1}^\top + \bar{u}u^\top - 2I = B^\dagger B - 2I$. The nonzero spectrum of $B^\dagger B$ is that of $BB^\dagger = \begin{pmatrix} N & \bar{f} \\ f & N \end{pmatrix}$. Verified against the explicitly enumerated line-graph adjacency at random $\theta$ (L11). □

This is the band-structure form of the ice rule: a configuration on the bonds whose sum vanishes on every simplex costs no hopping energy. For pyrochlore spins (which sit on the diamond bonds) this is spin ice's two-in–two-out constraint, and the frame's zero-sum hyperplane is literally its solution space at $\Gamma$.

**Theorem 9.2 (The pinch point is a cross-product square).** *Let $P(\theta)$ be the orthogonal projector onto the flat bands. As $\theta = \varepsilon\varphi \to 0$ with $\varphi \in H$,*

$$P(\varepsilon\varphi) \ \longrightarrow\ I - \tfrac{1}{N}J - \frac{\varphi\varphi^\top}{|\varphi|^2},$$

*which depends on the direction of approach (L11). For $N = 4$ this transverse projector equals*

$$-\frac{\widetilde{K}(\varphi)^2}{\varphi^\top M \varphi},$$

*the square of Paper 01's cross-product generator divided by the frame norm (exact, L11).*

*Proof.* $\ker B(\varepsilon\varphi) \to \{\sum_j v_j = 0,\ \sum_j \varphi_j v_j = 0\} = H \cap \varphi^\perp$. The cross-product form is Paper 01, Cor. 3.6 ($K(u)^2 = P_\parallel - I$ on $H$), rescaled to the $M$-normalization in which $\varphi^\top M \varphi = 4|\varphi|^2$. □

In the Gaussian (large-$\mathcal{N}$) treatment of the Coulomb phase, the spin correlations of pyrochlore spin ice are proportional to this flat-band projector, and its direction-dependent limit at $\Gamma$ is the dipolar pinch point seen in neutron scattering [Isakov et al., Henley]. Theorem 9.2 shows that this pinch point is the cross-product square of Paper 01. This is the doorway to the next target, spin ice treated natively in the frame.

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## 10. Scorecard: does the frame make the physics simpler?

**Pre-registered criteria.** A derivation is simpler if it uses fewer arbitrary choices, fewer irrational constants before the final conversion to physical units, fewer case splits, and if it makes symmetries manifest or covers several lattices with one argument. **Prediction:** the frame wins on lattices whose bond star is a regular simplex and gives nothing on lattices whose bond star is a cross-polytope.

| step (graphene, then diamond) | standard Cartesian derivation | frame derivation (this paper) |
|---|---|---|
| set-up | choose primitive vectors $a_1, a_2$ and a unit cell; $f = 1 + e^{ik\cdot a_1} + e^{ik\cdot a_2}$ | none: bonds = frame, $f = \sum_j e^{i\theta_j}$ (Prop. 6.3) |
| Brillouin zone | hexagon from $b_1, b_2$ with $\sqrt{3}$ factors | $T^N/U(1)$ (Prop. 6.2) |
| Dirac points | solve a two-variable trigonometric system; $K = \frac{2\pi}{3a}(1, \frac{1}{\sqrt{3}})$ | closing triangle; $\theta = 2\pi aj/3$, rational multiples of $\pi$ (Thm. 7.2) |
| valleys | $K$, $K'$ by inspection or time reversal | the two nontrivial $\mathbb{Z}/3$ characters, the two orientations |
| cone and isotropy | expansion with phase factors; isotropy checked afterwards | the genesis circle; isotropy forced by $S_3$ (Thm. 7.3) |
| Fermi velocity | $\frac{3}{2}ta$ | the tight-frame bound $N/(N-1)$ |
| diamond | a new calculation in the FCC zone | the same $f$ with $N = 4$; three matchings of $K_4$ (Thm. 7.4) |
| anisotropy | per-lattice analysis (Dirac merging; diamond studied numerically) | triangle and polygon inequalities; Grashof (Thm. 7.5); tropical diagram (Thm. 8.1) |

**Quantified reduction (graphene and diamond together).**

- *Arbitrary choices:* 2 per lattice (primitive cell, sublattice phase convention) against 0. The sublattice phase is the frame gauge, carried rather than fixed.
- *Irrational constants before the physical map:* at least 2 per lattice ($\sqrt{3}$ in $b_i$ and in $K$) against 0. The only irrational in the frame derivation is the final scale $\sqrt{N/(N-1)}$.
- *Case splits for the nodal set:* a two-variable trigonometric solve per lattice against one closing lemma per $N$, with the $N = 4$ lemma three lines long.
- *Marginal cost of the next lattice in the family:* a new calculation against zero.

**Control: the cubic lattice.** The band $2\sum_i \cos k_i$ separates in Cartesian coordinates because the bond star $\{\pm e_i\}$ is a cross-polytope: a product of $d$ copies of the $N = 2$ frame, which is the $(\mathbb{P}^1)^d$ side of Corollary 4.2. A single simplicial frame neither separates it nor removes any choice. **Confirmed: no gain.** The principle the test supports is that *a lattice's native frame is its bond star.* Cartesian coordinates are native to cross-polytope stars, the simplicial frame to simplex stars, and neither is better in general.

**Verdict.** On simplex lattices the physical results are unchanged and the derivations are measurably shorter and choice-free. Several facts known separately become one fact: Dirac points and valleys (characters and orientation), the diamond network (matchings of $K_4$), walk counts (Pearson moments), flat bands (zero-sum per simplex), and Kitaev's and the dimers' phase diagrams (amoeba and tropical line). The frame also produced three statements we did not find in the literature (Theorems 7.5, 8.1 and 9.2). It did not produce new physical predictions beyond Corollary 7.6, which is a consequence of the classical model made visible by the frame, not a new mechanism.

## 11. How to measure the Grashof loop count

Corollary 7.6 is the paper's one falsifiable prediction: a nearest-neighbour diamond-type band structure under [111] strain has **two** nodal loops under tension ($t' < 1$), **one** under compression ($1 < t' < 3$), and none beyond. The transition sits exactly at zero strain, where the isotropic network lies on all three Grashof walls at once. This section designs the measurement and simulates it, including its failure modes.

**Proposition 11.1 (Resolution requirement).** *Near a Grashof wall, with margin $g = (p+q)-(s+l)$, the two loops approach to a distance $\propto g^{1/2}$ in the space of phases, the square-root closing of a four-bar linkage near its change point. Numerically the separation is* 0.88, 0.63, 0.45 *and* 0.32 *rad at $g =$* 0.10, 0.05, 0.025 *and* 0.0125 *(fitted exponent* 0.49*; M1). A scan resolving about* 0.1 *rad counts the loops a few percent from the wall.*

**Route A: a synthetic Brillouin zone (fastest).** By Proposition 6.2 the Brillouin zone is $T^4/U(1)$ and the band is $|f|$ with $f = \sum_j t_j e^{i\theta_j}$. Any device that sums four controllable paths therefore reproduces the strained-diamond band structure exactly, and the unobservable global phase is the frame gauge. Candidate platforms:

- a four-arm optical interferometer (fiber or integrated) with set amplitude ratios $t_j$ and three phase shifters;
- a two-mode photonic coupler with four coupling channels;
- a qubit or two-level atom driven along four paths, whose Rabi frequency is $|f|$.

The output field $f(\theta)$ vanishes on the nodal set. Those zeros are the **vortex lines of the output phase** in the three-dimensional space of phase settings: the nodal loops are optical vortex loops, and the Grashof transition is a vortex reconnection.

**Proposition 11.2 (Counting by phase winding).** *Measure the complex output (heterodyne against a reference arm) on an $M^3$ grid of phase settings. Mark every grid plaquette around which* $\arg f$ *winds, connect the marked plaquettes into lines, and discard components shorter than $M/4$ plaquettes (noise rings). With $M = 64$ ($2.6 \times 10^5$ settings), 0.02 rad phase noise and 1% field noise:*

- *the count is* 2, 1, 0 *for $t' =$* 0.7, 1.3, 3.3 *in every repeat;*
- *it is correct in $\geq 90\%$ of repeats at $|g| = 2\%$ (M2).*

*The naive alternative, thresholding the dark-port intensity, fails: at the same resolution it merges two loops 10% from the wall into one (M3).*

**Route B: a real lattice.** Diamond networks with designed bond-dependent couplings are routine in acoustic metamaterials (coupled cavities, tube cross-section $\propto t_j$), microwave photonic crystals, and electric circuits (capacitive couplings, periodic boundary conditions with twisted phases). Cold atoms in a diamond optical lattice are possible but slower to realize. A bulk loop count needs three-dimensional momentum resolution; a surface measurement does not.

**Proposition 11.3 (The surface signature).** *On the* (111) *face — perpendicular to the strain axis — the zero-energy drumhead states fill the region of the surface Brillouin zone where $f$ winds along the surface normal. The complementary region (no zero mode) consists of* ***two pockets under tension and one under compression*** *($t' = 0.8, 0.9$ against $1.1, 1.2$). The total drumhead fraction varies smoothly across the wall (0.82 to 0.67 between $t' =$* 0.9 *and* 1.1*), so the measurable signature is the number of pockets in a zero-energy surface map, not a scalar (M4).*

**Scope.** Route A tests the identification of the nodal set with the planar quadrilateral space and the Grashof count on a synthetic zone. It is a clean, inexpensive test of Theorem 7.5 itself, but not of any material. Route B tests the crystal. Both assume nearest-neighbour bipartite hopping: a sublattice-symmetric next-neighbour term shifts the band energy without moving the band touching, whereas an A/B on-site energy difference gaps the loops and must be kept below the scan resolution.

## 12. Open problems and the next turn

- **F1 (Target 2: spin ice natively).** Carry the pyrochlore Coulomb phase fully into the frame: correlations from the projector of Theorem 9.2, emergent div from Paper 01b on the diamond's $K_4$ cells (Paper 01d), and monopoles as violations of the zero-sum (gauge) condition. Score it against Henley's Cartesian derivation. *(Carried out in Paper 01f, "Spin Ice in the Simplicial Frame": the lattice curl is $\widetilde{K}(e^{i\theta})$ and the photon obeys $\omega^2/\mathcal{UK} = 16 - |f|^2$.)*
- **F2 (Std-form limit shapes).** Lozenge tilings are the genesis cover image (Remark 4.4). Their limit shapes are governed by the Ronkin function of $f$ (Theorem 8.2) and freeze in the standard-form regions (Theorem 8.1). Write the arctic-curve theory in the two gauges.
- **F3 (Hyperdiamonds).** For $N \geq 5$ the nodal sets are planar polygon spaces of dimension $N-3$ with known Betti numbers [Farber–Schütz]. The isotropic Mahler measures $m(1 + x_1 + \cdots + x_{N-1})$ for $N = 5, 6$ have conjectured $L$-value expressions [Rodriguez-Villegas, Brunault–Zudilin]. Both are realizable in synthetic-dimension lattices.
- **F4 (Tensor products in the standard form).** Knutson–Tao honeycombs live on the genesis triangular grid. Read Littlewood–Richardson rules in the integer standard form of Proposition 4.5.
- **F5 (A piecewise-linear calculus).** The standard-form section has the piecewise-constant differential $\pi_{e_{\arg\min}}$ (Proposition 3.2). Develop the corresponding min-plus calculus (Hamilton–Jacobi and eikonal equations as $\tau \to 0$ limits of Paper 01b's heat equation) as the standard-form counterpart of Paper 01b.

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## Appendix A: Certificates

Deterministic, local, no GPU: `sympy`, `numpy`, `scipy`, `mpmath`.

`empirical_01e/cert_two_gauges.py` — **42/42.**

- **G1:** the metric for $N = 3$–$6$, gauge invariance, the genesis forms, and the refutation of the genesis two-zero formula.
- **G2:** linear gauges $\leftrightarrow w \cdot \mathbf{1} = 1$; barycentre and vertex gauges.
- **G3:** equivariance; gradient = Gibbs; $\tau = 1$ surprisal; cumulant expansion to third order; first correction = metric; the sandwich on 2000 random cases.
- **G4:** pairwise codimension-1 walls for $N = 3, 4, 5$; Cartesian adjacency counts for $d = 2, 3, 4$.
- **G5:** 0/1 vertices of the range ball and Voronoi = range ball for $A_2, A_3, A_4$ (3000 random points each).
- **G6:** $N$-ality, Young diagrams, massaction = removed columns, Weyl dimensions.

`empirical_01e/cert_simplex_lattices.py` — **44/44.**

- **L1:** closed walks on $D_3$ and $D_4$ = Brillouin-zone moments = $\sum$ multinomial$^2$; geometry.
- **L2:** $\theta = \sqrt{N/(N-1)}\, k$, zone = $T^N/U(1)$.
- **L3:** $|f|^2$ as an XY energy; gauge = Bloch phase; spectrum.
- **L4:** exact Dirac points, characters, $|\Gamma K|$.
- **L5:** the cone = genesis circle, $\frac{3}{2}(q_x + iq_y)$, $v_F$.
- **L6:** the pairing lemma (3000 random nodal points), three circles, crossings at $X$, the circle's radial range $[|\Gamma X|, |\Gamma W|]$.
- **L7:** the law of cosines and the triangle criterion (4000 random cases).
- **L8:** four-bar count = voxel count ($448^3$) = Grashof on 56/56; strain examples.
- **L9:** walls $\subseteq G_\tau \subseteq \{\delta \leq \tau \log(N-1)\}$ (20000 cases); sharpness for $N = 3$–$8$; Kitaev's medial triangle.
- **L10:** $m(1+x+y)$ to 25 digits, $m(1+x+y+z)$ to 9 digits, Ronkin equality on closed gapped phases, the sandwich.

- **L11:** kagome and pyrochlore $H = B^\dagger B - 2$ from explicit adjacency; flat space at $\Gamma = H$; pinch-point limit; $-\widetilde{K}(\varphi)^2/\varphi^\top M\varphi$ exactly.

Transcripts: `cert_two_gauges.out.txt`, `cert_simplex_lattices.out.txt`.

**Recorded failure.** The first loop counter for L8 traced nodal points along one parametrization and joined them by proximity. It disagreed with the four-bar count on most cases, because near a fold the two branches separate like $\sqrt{\cdot}$ and proximity joining misses the connection. It was replaced by the independent voxel count. That count at $192^3$ still merged distinct loops near the Grashof wall (50/56), and it converged at $448^3$ (56/56) to the count predicted before the run. The theorem was not adjusted to fit the numerics.

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`empirical_01e/cert_grashof_measurement.py` — **6/6** (Section 11).

- **M1:** loop separation $\propto$ margin$^{1/2}$.
- **M2:** vortex counting on the synthetic zone, [111]-strain counts 2/1/0, and $\geq$ 90% success at 2% margin.
- **M3:** the intensity-threshold failure mode.
- **M4:** (111) surface pockets 2 against 1, with a smooth drumhead fraction.

**Recorded failure (Section 11).** The first loop-counting design thresholded the dark-port intensity and merged the two loops even at a 10% margin. Vortex counting replaced it. That counter's first version itself got worse with finer grids, because noise creates small vortex rings; a size filter fixed it.

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*[End of Draft 1.]*